where $P$ is the number of programmes printed and $D$ is the demand for programmes.

(b) largest expected profit = Rs 2,260 accrues for $P$ = 30,000 copies per game

(c) In case of perfect forecast, the expected profit is:

$900 \times 0.1 + 2{,}000 \times 0.4 + 3{,}100 \times 0.3 + 4{,}200 \times 0.2 = \text{Rs } 2{,}660.$

EVPI = Rs 2,660 – Rs 2,260 = Rs 400

10. Cost function $= \begin{cases} 70\,(D - S) & ;\ D \geq S \\ 200\,(S - D) & ;\ D < S \end{cases}$

where $D$ = monthly demand (or sales); $S$ = number of units purchased

Since the expected cost of Rs 46 is minimum for course of action 4, the optimum stock to minimize the cost is 4 units per month.

11. Cost function $= \begin{cases} 500\,(D - S) & ;\ D \geq S \\ 200\,(S - D) & ;\ D < S \end{cases}$

Since the expected cost of Rs 315 is minimum for course of action 4, the optimum stock to minimize the cost is 4 cooking ranges.

12. (b) small stockpile ; Rs 30,000 – Rs 11,000 ; (d) Rs 19,000

13. Marginal profit (MP) = Rs (35 – 15) = Rs 20

Marginal loss (ML) = Rs (15 – 10) = Rs 5

Conditional payoff = MP × garments sold – ML × garments unsold

(a) EMV ($S_1$) = Rs 600; EMV ($S_2$) = Rs 775;
EMV ($S_3$) = Rs 887.5;
EMV ($S_4$) = Rs 900; EMV ($S_5$) = Rs 850
Max. EMV is corresponding to course of action $S_4$, i.e. purchase 60 garments.

(b) EMV ($S_2$) = Rs 800; EMV ($S_3$) = Rs 962.5,
EMV ($S_4$) = Rs 1,037.5;
EMV ($S_5$) = Rs 1,012.5.
Max. EMV is corresponding to course of action $S_4$, i.e. purchase 60 garments.

14. (a) First alternative, Rs 20,000

(b) EVPI = Expected cost under uncertainty
– Expected cost with perfect information
$= 20{,}000 - (16{,}000 \times 0.2 + 20{,}000 \times 0.7 + 20{,}000 \times 0.1)$
$= 20{,}000 - 19{,}200 = \text{Rs } 800$ per batch.

## 11.7 DECISION TREE ANALYSIS

Decision-making problems discussed earlier were limited to arrive at a decision over a fixed period of time. That is, payoffs, states of nature, courses of action and probabilities associated with the occurrence of states of nature were not subject to change.

However, situations may arise when a decision-maker needs to revise his previous decisions due to availability of additional information. Thus he intends to make a sequence of interrelated decisions over several future periods. Such a situation is called a *sequential or multiperiod decision process*. For example, in the process of marketing a new product, a company usually first go for 'Test Marketing' and other alternative courses of action might be either 'Intensive Testing' or 'Gradual Testing'. Given the various possible consequences – good, fair, or poor, the company may be required to decide between redesigning the product, an aggressive advertising campaign or complete withdrawal of product, etc. Based on this decision there might be an outcome that leads to another decision and so on.

A decision tree analysis involves the construction of a diagram that shows, at a glance, when decisions are expected to be made – in what sequence, their possible outcomes, and the corresponding payoffs.

**Decision tree** is the graphical display of the progression of decision and random events.

A decision tree consists of *nodes, branches, probability estimates,* and *payoffs*. There are two types of nodes:

- ***Decision (or act) node:*** A decision node is represented by a square and represents a point of time where a decision-maker must select one *alternative course of action* among the available. The courses of action are shown as *branches* or *arcs* emerging out of decision node.
- ***Chance (or event) node:*** Each course of action may result in a *chance node*. The chance node is represented by a circle and indicates a point of time where the decision-maker will discover the response to his decision.

*Branches* emerge from and connect various nodes and represent either decisions or states of nature. There are two types of branches:

- ***Decision branch:*** It is the branch leading away from a decision node and represents a course of action that can be chosen at a decision point.
- ***Chance branch:*** It is the branch leading away from a chance node and represents the state of nature of a set of chance events. The assumed probabilities of the states of nature are written alongside their respective chance branch.

- ***Terminal branch:*** Any branch that makes the end of the decision tree (not followed by either a decision or chance node), is called a *terminal branch.* A terminal branch can represent either a course of action. The terminal points of a decision tree are supposed to be mutually exclusive points so that exactly one course of action will be chosen.

The *payoff* can be positive (i.e. revenue or sales) or negative (i.e. expenditure or cost) and it can be associated either with decision or chance branches.

An illustration of a decision tree is shown in Fig. 11.2. It is possible for a decision tree to be deterministic or probabilistic. It can also further be divided in terms of stages – into single stage (a decision under condition of certainty) and multistage (a sequence of decisions).

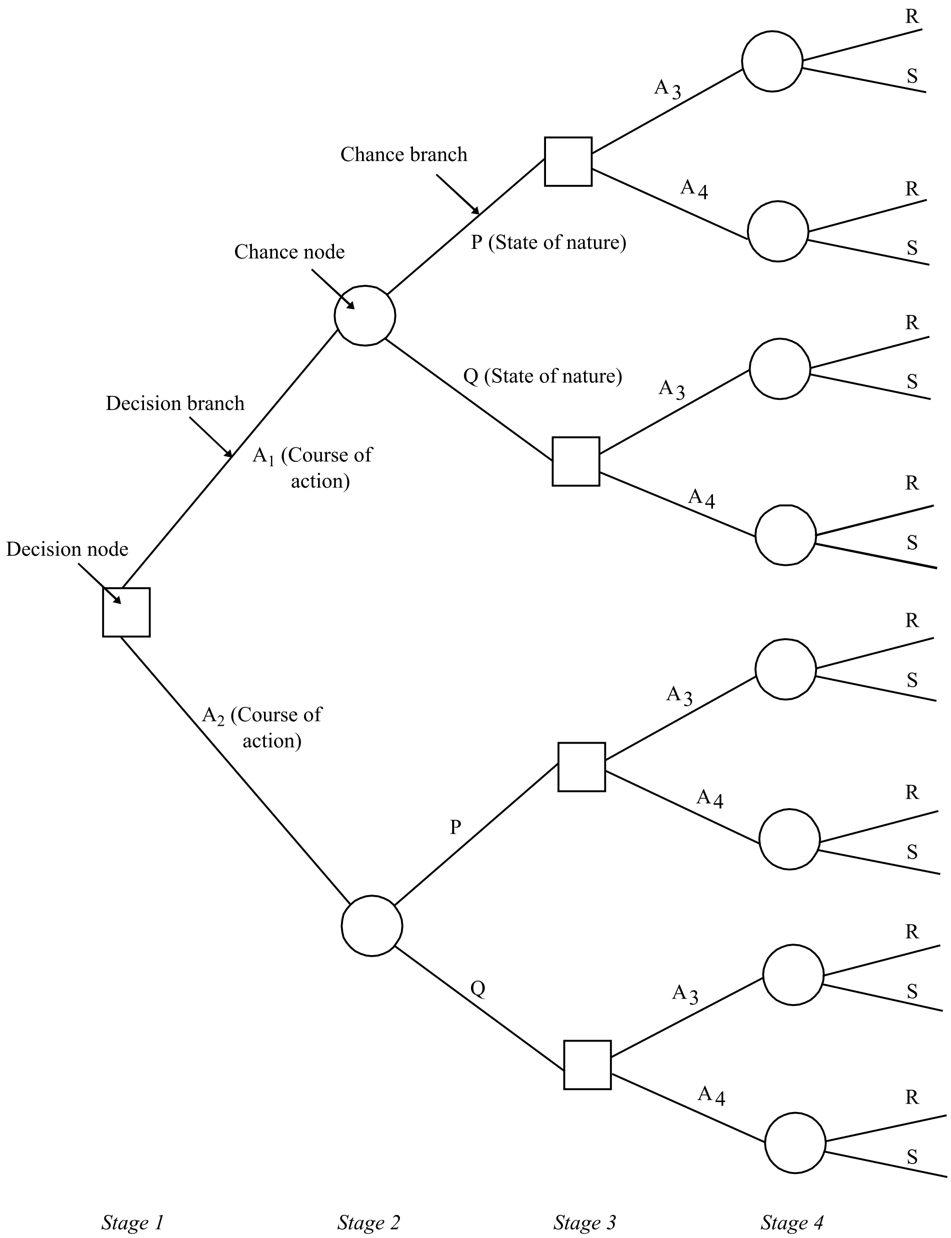


**Fig. 11.2**
Decision Tree

The optimal sequence of decisions in a tree is found by starting at the right-hand side and rolling backwards. At each node, an expected return is calculated (called *position value*). If the node is a chance node, then the position value is calculated as the sum of the products of the probabilities or the branches emanating from the chance node and their respective position values. If the node is a decision node, then the expected return is calculated for each of its branches and the highest return is selected. This procedure continues until the initial node is reached. The position values for this node corresponds to the maximum expected return obtainable from the decision sequence.

**Remark** *Decision trees versus probability trees* Decision trees are basically an extension of probability trees. However, there are several basic differences:

(i) The decision tree utilizes the concept of 'rollback' to solve a problem. This means that it starts at the right-hand terminus with the highest expected value of the tree and works back to the current or beginning decision point in order to determine the decision or decisions that should be made. It is the multiplicity of decision points that make the rollback process necessary.

(ii) The probability tree is primarily concerned with calculating the probabilities, whereas the decision tree utilizes probability factors as a means of arriving at a final answer.

(iii) The most important feature of the decision tree, is that it takes time differences of future earnings into account. At any stage of the decision tree, it may be necessary to weigh differences in immediate cost or revenue against differences in value at the next stage.

**Example 11.20** You are given the following estimates concerning a Research and Development programme:

| *Decision* $D_i$ | *Probability of Decision* $D_i$ *Given Research R* $P(D_i \mid R)$ | *Outcome Number* | *Probability of Outcome* $x_i$ *Given* $D_i$ $P(x_i \mid D_i)$ | *Payoff Value of Outcome,* $x_i$ (*Rs '000*) |
|---|---|---|---|---|
| Develop | 0.5 | 1 | 0.6 | 600 |
| | | 2 | 0.3 | – 100 |
| | | 3 | 0.1 | 0 |
| Do not develop | 0.5 | 1 | 0.0 | 600 |
| | | 2 | 0.0 | – 100 |
| | | 3 | 1.0 | 0 |

Construct and evaluate the decision tree diagram for the above data. Show your workings for evaluation.

**Solution** The decision tree of the given problem along with necessary calculations is shown in Fig. 11.3.

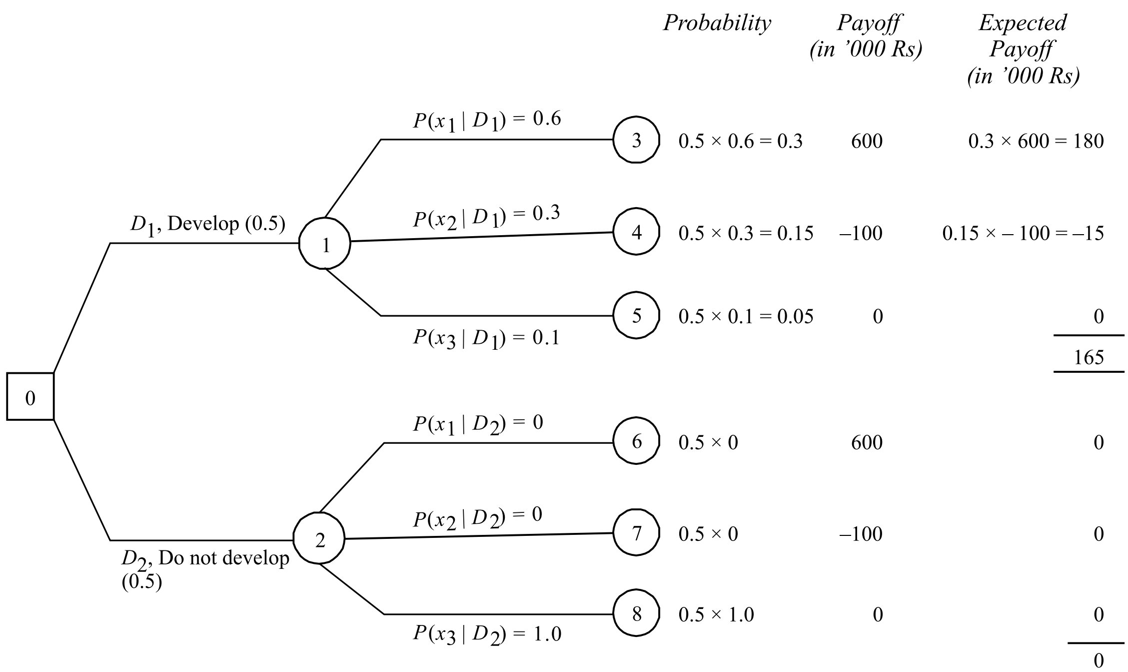


**Fig. 11.3** Decision Tree

**Example 11.21** A glass factory that specializes in crystal is developing a substantial backlog and for this the firm's management is considering three courses of action: To arrange for subcontracting ($S_1$), to begin overtime production ($S_2$), and to construct new facilities ($S_3$). The correct choice depends largely upon the future demand, which may be low, medium, or high. By consensus, management ranks the respective probabilities as 0.10, 0.50 and 0.40. A cost analysis reveals the effect upon the profits. This is shown in the table below:

| *Demand* | *Probability* | *Course of Action* | | |
|---|---|---|---|---|
| | | $S_1$ (*Subcontracting*) | $S_2$ (*Begin Overtime*) | $S_3$ (*Construct Facilities*) |
| Low (L) | 0.10 | 10 | – 20 | – 150 |
| Medium (M) | 0.50 | 50 | 60 | 20 |
| High (H) | 0.40 | 50 | 100 | 200 |

Show this decision situation in the form of a decision tree and indicate the most preferred decision and its corresponding expected value.

**Solution** A decision tree that represents possible courses of action and states of nature is shown in Fig. 11.4. In order to analyze the tree, we start working backwards from the end branches.

The most preferred decision at the decision node 0 is found by calculating the expected value of each decision branch and selecting the path (course of action) that has the highest value.

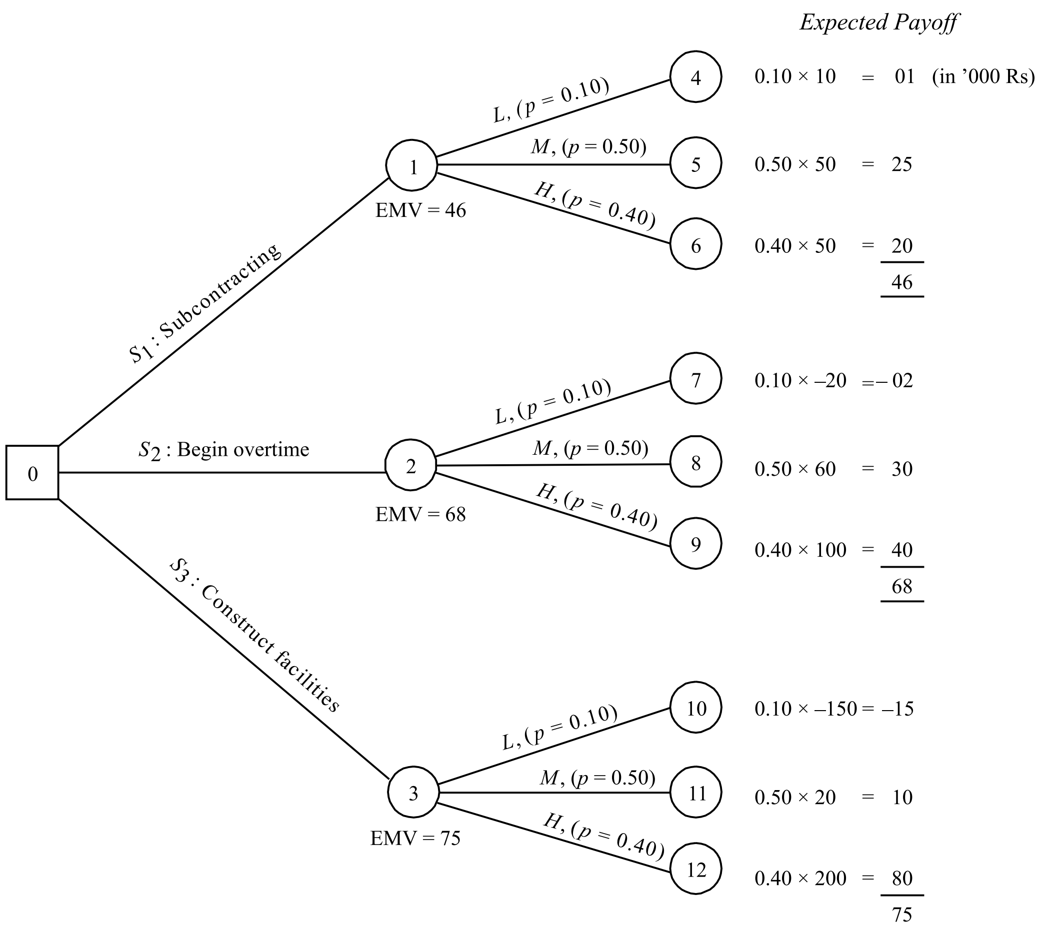


**Fig. 11.4**
Decision Tree

Since node 3 has the highest EMV, therefore, the decision at node 0 will be to choose the course of action $S_3$, i.e. construct new facilities.

**Example 11.22** A businessman has two independent investment portfolios A and B, available to him, but he lacks the capital to undertake both of them simultaneously. He can either choose A first and then stop, or if A is not successful, then take, B or vice versa. The probability of success of A is 0.6, while for B it is 0.4. Both investment schemes require an initial capital outlay of Rs 10,000 and both return nothing if the venture proves to be unsuccessful. Successful completion of A will return Rs 20,000 (over cost) and successful completion of B will return Rs 24,000 (over cost). Draw a decision tree in order to determine the best strategy. [*Delhi Univ., MBA, 2000, AMIE, 2006*]

**Solution** The decision tree based on the given information is shown in Fig. 11.5. The evaluation of each chance node and decision is given in Table 11.33.

| Decision Point | | Outcome | Probability | Conditional Value (*Rs*) | Expected Value |
|---|---|---|---|---|---|
| $D_3$ | (i) Accept *A* | Success | 0.6 | 20,000 | 12,000 |
| | | Failure | 0.4 | –10,000 | – 4,000 |
| | | | | | 8,000 |
| | (ii) Stop | – | – | – | 0 |
| $D_2$ | (i) Accept *B* | Success | 0.4 | 24,000 | 9,600 |
| | | Failure | 0.6 | –10,000 | – 6,000 |
| | | | | | 3,600 |
| | (ii) Stop | – | – | – | 0 |
| $D_1$ | (i) Accept *A* | Success | 0.6 | 20,000 + 3,600 = 23,600 | 14,160 |
| | | Failure | 0.4 | –10,000 | – 4,000 |
| | | | | | 10,160 |
| | (ii) Accept *B* | Success | 0.4 | 24,000 + 8,000 = 32,000 | 12,800 |
| | | Failure | 0.6 | –10,000 | – 6,000 |
| | | | | | 6,800 |
| | (iii) Do nothing | – | – | – | 0 |

**Table 11.33** Evaluation of Decision and Chance Nodes

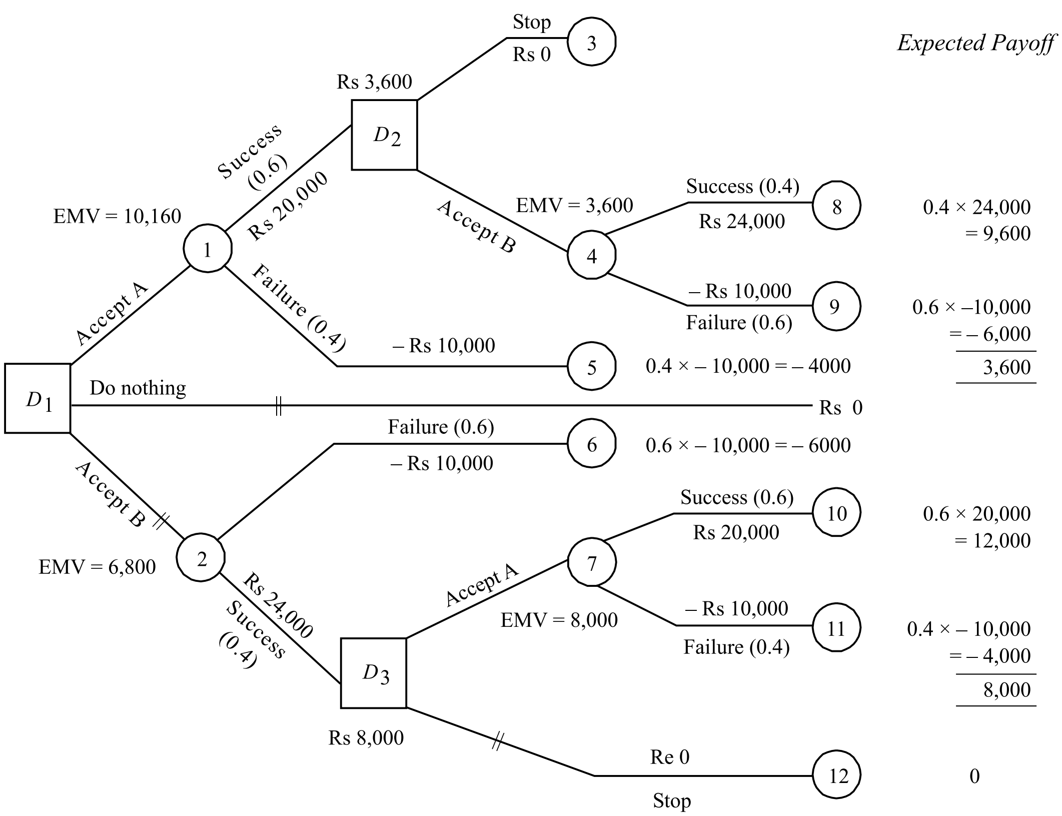


**Fig. 11.5** Decision Tree

Since the EMV = Rs 10,160 at node $D_1$ is highest, therefore the best strategy is to accept course of action A first and if A is successful, then accept B.

**Example 11.23** The Oil India Corporation (OIC) is wondering whether to go for an offshore oil drilling contract that is to be awarded in Bombay High. If OIC bid, value would be Rs 600 million with a 65 per cent chance of gaining the contract. The OIC may set up a new drilling operation or move the already existing operation, which has already proved successful for a new site. The probability of success and expected returns are as follows:

| *Outcome* | *New Drilling Operation* | | *Existing Operation* | |
|---|---|---|---|---|
| | *Probability* | *Expected Revenue* (*Rs million*) | *Probability* | *Expected Revenue* (*Rs million*) |
| Success | 0.75 | 800 | 0.85 | 700 |
| Failure | 0.25 | 200 | 0.15 | 350 |

If the Corporation do not bid or lose the contract, they can use Rs 600 million to modernize their operation. This would result in a return of either 5 per cent or 8 per cent on the sum invested with probabilities 0.45 and 0.55. (Assume that all costs and revenue have been discounted to present value.)

(a) Construct a decision tree for the problem showing clearly the courses of action.

(b) By applying an appropriate decision criterion recommend whether or not the Oil India Corporation should bid the contract. [*Delhi Univ. MBA, AMIE, 2001, 2005*]

**Solution** The decision tree based on the given information is shown in Fig. 11.6. The evaluation of each chance node and decision node is given in Table 11.34.

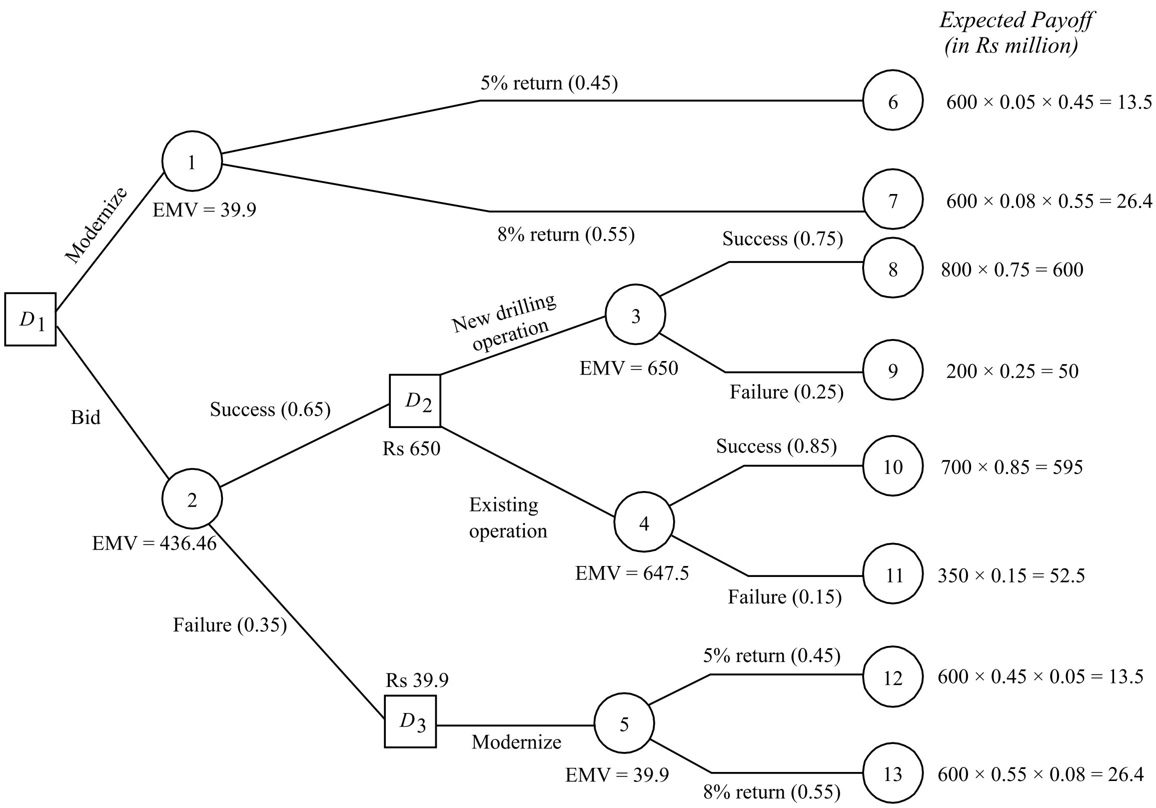


**Fig. 11.6**
Decision Tree

| Decision Point | | Outcome | Probability | Conditional Value | Expected Value (Rs) |
|---|---|---|---|---|---|
| $D_3$ | (i) Modernize | 5% return | 0.45 | 600 × 0.05 = 30 | 30 × 0.45 = 13.5 |
| | | 8% return | 0.55 | 600 × 0.08 = 48 | 48 × 0.55 = 26.4 |
| | | | | | 39.9 |
| $D_2$ | (i) Undertake new | Success | 0.75 | 800 | 600 |
| | drilling operation | Failure | 0.25 | 200 | 50 |
| | | | | | 650 |
| | (ii) Move existing | Success | 0.85 | 700 | 595 |
| | operation | Failure | 0.15 | 350 | 52.5 |
| | | | | | 647.5 |
| $D_1$ | (i) Modernize | 5% return | 0.45 | 600 × 0.05 = 30 | 30 × 0.45 = 13.5 |
| | | 8% return | 0.55 | 600 × 0.08 = 48 | 48 × 0.55 = 26.4 |
| | | | | | 39.9 |
| | (ii) Bid | Success | 0.65 | 650 | 422.50 |
| | | Failure | 0.35 | 39.9 | 13.96 |
| | | | | | 436.46 |

**Table 11.34** Evaluation of Decision and Chance Nodes

Since EMV, Rs 436.46 at event node 2 is highest, therefore the best decision at decision node $D_1$ is to decide for bid and if successful establish a new drilling operation.

**Example 11.24** A large steel manufacturing company has three options with regard to production: (*i*) produce commercially (*ii*) build pilot plant (*iii*) stop producing steel. The management has estimated that their pilot plant, if built, has 0.8 chance of high yield and 0.2 chance of low yield. If the pilot plant does show a hight yield, management assigns a probability of 0.75 that the commercial plant will also have a high yield. If the pilot plant shows a low yield, there is only a 0.1 chance that the commercial plant will show a high yield. Finally, management's best assessment of the yield on a commercial-size plant without building a pilot plant first has a 0.6 chance of high yield. A pilot plant will cost Rs. 3,00,000. The profits earned under high and low yield conditions are Rs. 1,20,00,000 and – Rs. 12,00,000 respectively. Find the optimum decision for the company. *[Punjab Tech Univ., BE, 2006]*

**Solution** A decision tree representing possible courses of action and states of nature are shown in Fig. 11.7. In order to analyse the tree, we proceed backward from the end branches

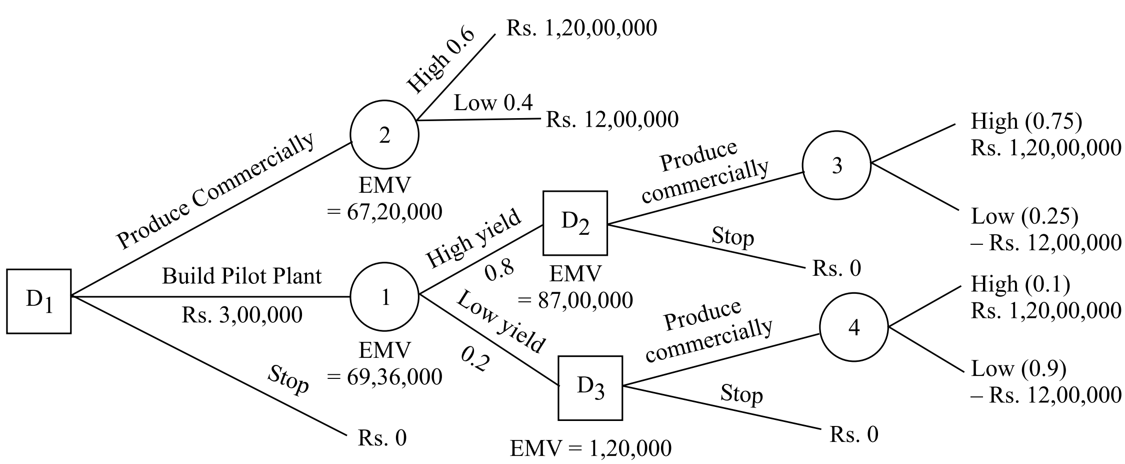


**Fig. 11.7**

$$\text{EMV (Node 3)} = 0.75 \times 1{,}20{,}00{,}000 - 0.25 \times 12{,}00{,}000 = \text{Rs. } 87{,}00{,}000$$

$$\text{EMV (Node 4)} = 0.1 \times 1{,}20{,}00{,}000 - 0.9 \times 12{,}00{,}000 = 12{,}00{,}000 - 10{,}80{,}000 = \text{Rs. } 1{,}20{,}000.$$

$$\text{EMV (Node 1)} = 0.8 \times 87{,}00{,}000 - 0.2 \times 1{,}20{,}000 = \text{Rs. } 69{,}36{,}000$$

$$\text{EMV (Node 2)} = 0.6 \times 1{,}20{,}00{,}000 - 0.4 \times 12{,}00{,}000 = 72{,}00{,}000 - 4{,}800{,}000 = \text{Rs. } 67{,}20{,}000$$

$$\text{EMV (Node D}_2) = \text{Rs. } 87{,}00{,}000$$

$$\text{EMV (Node D}_3) = 1{,}20{,}000.$$

$$\text{EMV (Node D}_1) = 69{,}36{,}000 - 3{,}00{,}000 = \text{Rs. } 66{,}36{,}000.$$

Since at decision node $D_1$ the production cost of Rs 67,20,000, associated with course of action – Build pilot plant is least, the company should build pilot plant.

**Example 11.25** A businessman has an option of selling a product in domestic market or in export market. The available relevant data are given below.

| *Items* | *Export Market* | *Domestic Market* |
|---|---|---|
| Probability of selling | 0.6 | 1.0 |
| Probability of keeping delivery schedule | 0.8 | 0.9 |
| Penalty of not meeting delivery schedule (Rs.) | 50,000 | 10,000 |
| Selling price (Rs.) | 9,00,000 | 8,00,000 |
| Cost of third party inspection (Rs.) | 30,000 | Nil |
| Probability of collection of sale amount | 0.9 | 0.9 |

If the product is not sold in export market, it can always be sold in domestic market. There are no other implications like interest and time.

(a) Draw the decision tree using the data given above.
(b) Should the businessman go for selling the product in the export market? Justify your answer.

**Solution** (i) The decision tree representing possible curses of action and states of nature is shown in Fig. 11.8.

The revenue generated from the stage of product in export market is equal to the selling price minus inspection cost (i.e. Rs. 30,000). The tree is analyzed by moving backward from the end branches.

$$\text{EMV (Node 9)} = \text{Rs. } 0.9 \times 8{,}00{,}000 = 0.9 \times 8{,}00{,}000 = \text{Rs. } 7{,}20{,}000.$$

$$\text{EMV (Node 10)} = \text{Rs. } 7{,}20{,}000.$$

$$\text{EMV (Node 7)} = \text{Rs. } 0.9 \times 8{,}70{,}000 - 0.1 \times 30{,}000 = \text{Rs. } 7{,}80{,}000.$$

$$\text{EMV (Node 8)} = \text{Rs. } 7{,}80{,}000.$$

$$\text{EMV (Node 4)} = 0.9 \times 7{,}20{,}000 + 0.1 \times 7{,}10{,}000 = \text{Rs. } 7{,}19{,}000.$$

$$\text{EMV (Node 3)} = 0.8 \times 7{,}80{,}000 + 0.2 \times 7{,}30{,}000 = \text{Rs. } 7{,}70{,}000.$$

$$\text{EMV (Node 5)} = 0.9 \times 8{,}00{,}000 = \text{Rs. } 72{,}20{,}000.$$

$$\text{EMV (Node 6)} = 7{,}20{,}00.$$

$$\text{EMV (Node 1)} = 0.6 \times 7{,}70{,}000 + 0.4 \times 7{,}19{,}000 = \text{Rs. } 7{,}49{,}600.$$

$$\text{EMV (Node 2)} = 0.9 \times 7{,}20{,}000 + 0.1 \times 7{,}10{,}000 = \text{Rs. } 7{,}19{,}000.$$

$$\text{EMV (Node D}_1) = \text{Max. } \{7{,}49{,}600;\ 7{,}19{,}000\} = \text{Rs. } 7{,}49{,}600.$$

Hence, the businessman should go for selling the product in the export market.

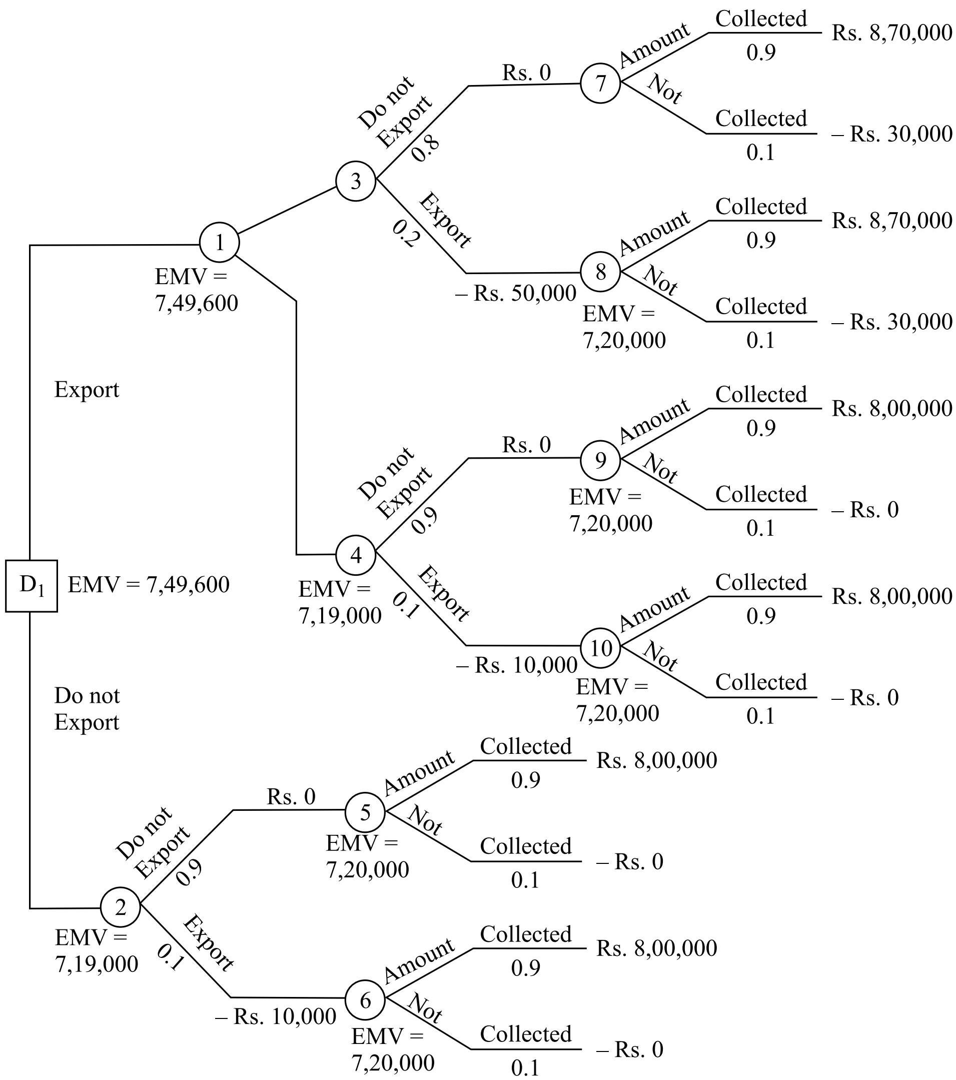


**Fig. 11.8**

## 11.8 DECISION-MAKING WITH UTILITIES

In previous sections, we discussed solution to problems based on the criterion of expected profit (or loss) expressed in monetary terms. However, in many situations, such criterion that involves expected monetary payoff may not be appropriate. This is because of the fact that different individuals attach different utility to money, under different conditions. The term utility is *the measure of preference for various alternatives in terms of relative value for money.* The utility of a given alternative is unique to the individual decision-makers and unlike a simple monetary amount, can incorporate intangible factors or subjective standards from their own value systems.

**Illustration** Mr Ram has won Rs 1,000 in a quiz programme. In the last round he is asked either to compete or quit. Now he has two alternatives: (a) quit and take his winnings, (b) take a last chance in which he has 50–50 chance of winning Rs 4,000 or nothing. The question now is: What would he do? On EMV basis he has

$$\text{EMV } (b) = 0.50\ (4,000) + 0.50\ (0) = \text{Rs } 2,000$$

**Utility** provides a way of incorporating the decision maker's attitude towards risk in arriving at a decision and hence measures the true value of an outcome.

This amount is twice what he has already won. But would he really give up Rs 1,000 for a 50–50 chance or Rs 4,000 or nothing? Many individuals would not because they would think of all the alternatives they could do with Rs 1,000 and how they would regret it if they end up with nothing. Hence a new payoff measure *utility* reflecting the decision-maker's attitude and preference has to be introduced.

The basic axioms of utility may be stated as follows:

(i) If outcome $A$ is preferred to outcome $B$, then the utility $U(A)$ of outcome $A$ is greater than the utility $U(B)$ of outcome $B$, and vice versa. If both are equally preferred, then $U(A) = U(B)$

(ii) If the decision-maker does not see difference between the two alternatives and outcome $A$ occured with probability $p$, and outcome $C$ with probability $(1 - p)$, then $U(B) = pU(A) + (1 - p)\, U(C)$

Under utility criterion, a rational decision-maker will choose an alternative that optimizes the *expected utility* rather than expected monetary value. Once individual's utility function is known, along with the probability assigned to outcomes, then the total expected utility for each course of action can be obtained by multiplying the utility values with their probabilities. The strategy that corresponds to the optimum utility function is called the *equal strategy*.

### 11.8.1 Utility Functions

Utility function describes the relative preference value that individuals have for a given criterion such as money, goods, etc. Preferences are often determined by choosing between receiving a given amount, for a certain activity versus a 50–50 chance of gaining a larger amount or nothing. The gamble amount is adjusted upward or downward until the individual is indifferent to whether he receives the certain amount or the gamble. This indifference point establishes individual's utility.

A utility function can be used to convert a decision criteria value into *utils* so that a decision can be made on the basis of maximizing the expected utility value (EUV) rather than, say the EMV.

### 11.8.2 Utility Curve

**Utility function** is a graphically display of utility values – amount of money goods, etc. received.

A utility curve that relates utility values to rupee values is obtained by placing the decision-maker in various hypothetical decision situations and plotting the decision-maker's pattern of choices in terms of risk and utilities. Fig. 11.9 shows several utility curves and the risk preference associated with each.

Suppose, the relationship between monetary gains, losses and utilities for gains is established. As shown in Fig. 11.9, if the utility curve is bent down non-linearly, then a large negative utility is assigned for losses. It is important not to make the curve bend down too steeply or to start the bending too quickly because this may lead individuals into a situation where they attach such a heavy weightage to the possibility of loss that they never take any risk and thus, never make any gains.

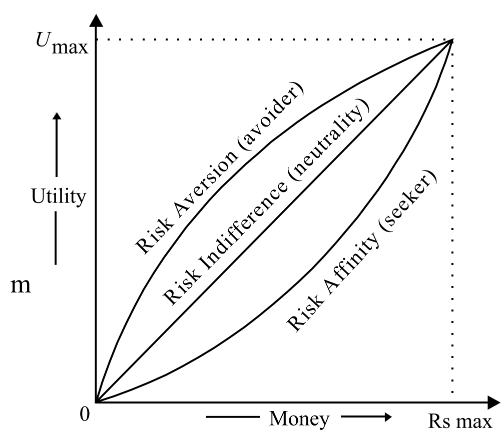


**Fig. 11.9** Utility Curve and Associated Risk Preferences

On the positive side of the curve, it eventually bend away from the straight line. This indicates that increasing units of money are resulting in smaller additional gains in utility.

### 11.8.3 Construction of Utility Curves

Suppose, there is a project that would payoff with Rs 1000 or (–) Rs 1000. Then what probability of success would make the decision-maker just about indifferent as to whether to undertake the project or not? Suppose he says that he would require an 80 per cent probability of success to make him indifferent about undertaking the project. Suppose utility of Rs 1000 is 10 for an individual. Then

$$U(\text{Re } 0) = (0.80)\, U(\text{Rs } 1000) + (0.20)\, U(-\text{ Rs } 1000)$$
$$0 = 0.80\ (10) + (0.20)\ U(-\text{ Rs } 10) \text{ or } U(-\text{ Rs } 10) = -40$$

The utility curve will have three points:

$$U(+1000) = 10; \quad U(0) = 0 \quad \text{and} \quad U(-1000) = -40$$

Other points on the curve can be obtained by interacting with decision-maker to know his/her utility values to a payoff.

**Example 11.27** A manager must choose between two investments A and B that are calculated to yield net profits of Rs 1,200 and Rs 1,600, respectively with probabilities subjectively estimated at 0.75 and 0.60. Assume, the manager's utility function reveals that utilities for Rs 1,200 and Rs 1,600 are 45 and 50 utils, respectively. What is the best choice on the basis of the expected utility value (EUV)?

**Solution** The expected utility value is expressed as:

$$\text{EUV} = \sum_{i=1}^{m} U_i p_i$$

where $U_i$ = utility value of state of nature $i$

$p_i$ = probability value of state of nature $i$

Then EUV $(A) = p_A U_A = 0.75 \times 45 = 33.75$ utils and EUV $(B) = p_B U_B = 0.60 \times 50 = 30$ utils

Since EUV($A$) > EUV($B$), therefore, the best choice is investment $A$.

**Example 11.28** A businessman is thinking whether to invest in bonds with an assured return or in a new venture that is likely to fetch him Rs. 20,000 or nothing with equal probabilities. The businessman says he would prefer an assured return in it is Rs. 10,000 or more, would be indifferent to the two alternatives if the assured return is Rs. 8,000 and would opt for the risky alternative if this amount is less than Rs. 6,000.

The businessman had purchased a minicomputer last year which is lying with him unused at present. The minicomputer can fetch him a profit of Rs. 8,000 by way of consultancy. He says that he would not like to sell the computer but would be indifferent to an offer of Rs. 3,000.

Recently a Govt agency invited bid offers for a contract worth a profit of Rs. 8,000. Bidding expenses are reimbursable if the contract materializes and not otherwise. There is an equal chance of winning or losing the bid. After thinking over the possibilities, the businessman says that he would be indifferent to submitting the bid at a bidding expense of Rs. 2,000. Construct the businessman's utility curve by using the CME technique.

**Solution** Let arbitrary utility values of 100 and 0 be assigned to the most favourable and least favourable outcomes, i.e.

$$U_{20,000} = 100 \text{ utiles}, \quad U_0 = 0 \text{ utile.}$$

Given that $U_{8,000} = 0.5\ U_{20,000} + 0.5\ U_0 = 0.5 \times 100 + 0.5 \times 0 = 25$ utiles.

The businessman is indifferent between his utilities for Rs. 3,000 and of a chance of Rs. 8,000 or nothing. Thus,

$$U_{3,000} = 0.5\ U_{8,000} + 0.5\ U_0 = 0.5 \times 50 + 0.5 \times 0 = 25 \text{ utiles.}$$

The decision to enter the bid at an expense of Rs. 2,000 may get him either a gain of Rs. 8,000 or lose (bidding expenses of) Rs. 2,000 with equal probability. Since he is indifferent to submitting the bid,

$$U_0 = 0.5\, U_{8,000} + 0.5\, U_{-2,000}$$
$$0 = 0.5 \times 50 + 0.5\, U_{-2,000} \quad \text{or} \quad U_{-2,000} = -50 \text{ utiles.}$$

Following in the summary of points for the businessman's utility curve:

| *Amount (Rs.)* | *Utility (utiles)* |
|---|---|
| 20,000 | 100 |
| 8,000 | 50 |
| 3,000 | 25 |
| 0 | 0 |
| – 2,000 | – 50 |

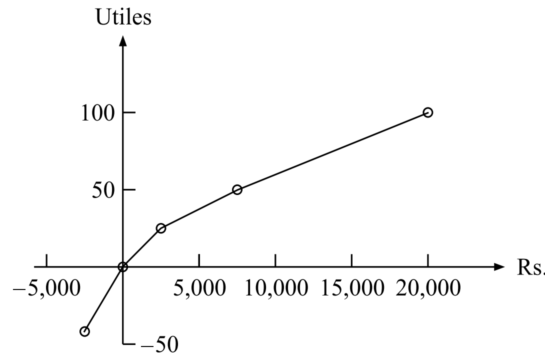


**Fig. 11.10** Utility Curve

The utility curve is plotted in Fig. 11.10. This curve may reflect a reasonable estimate of the businessman's behaviour when faced with risky ventures.

**Example 11.29** Mr XX has an after-tax annual of Rs. 90,000 and is considering to buy accident insurance for his car. The probability of accident during the year is 0.1 (assume that at most one accident will occur), in which case the damage to the car will be Rs. 11,600. With a utility function of $U(x) = \sqrt{x}$, what is the insurance premium he will be willing to pay?

**Solution** Let $A$ represent the venture when Mr XX does not by the accident insurance for his car. Then, in case of accident he would spend Rs. 11,600 on damages and will be left with Rs. 78,400. In case of no accident he retains Rs. 90,000. Then we have

$$U_A = U_{78,400} \times 0.1 + U_{90,000} \times 0.9.$$

Since $U(x) = U_x = \sqrt{x}$, $U_{78,400} = \sqrt{78,400} = 280$ utiles, and $U_{90,000} = \sqrt{90,000} = 300$ utiles. Thus, $U_A = 280 \times 0.1 + 300 \times 0.9 = 298$ utiles.

So the amount Rs. $x$ which will give the same utility of the venture: $A = (298)^2 =$ Rs. 88,804. [Because $U(X) = \sqrt{x}$ or $X = [U(x)]^2$]

Thus Mr XX will be indifferent to an amount of Rs. 88,804 with certainty and the venture $A$. The amount he will be willing to pay as car premium would be 90,000 – 88,804 = Rs. 1,196.

## SELF PRACTICE PROBLEMS C

1. Raman Industries Ltd. has a new product that they expect has great potential. At the moment they have two courses of action open to them: To test market ($S_1$) and to drop product ($S_2$).
   If they test it, it will cost them Rs 50,000. The response could be either positive or negative, with probabilities 0.70 and 0.30, respectively. If it is positive, they could either market it with full effort or drop the product. If they market with full scale, then the result might be low, medium, or high demand and the respective net payoffs would be – Rs 1,00,000, Rs 1,00,000 or Rs 5,00,000. These outcomes have probabilities of 0.25, 0.55 and 0.20, respectively. If the result of the test marketing is negative they would decide to drop the product. If, at any point, they drop the product there is a net gain of Rs 25,000 from the sale of scrap. All financial values have been discounted to the present. Draw a decision tree for the problem and indicate the most preferred decision.
2. A manufacturing company has just developed a new product. On the basis of past experience, a product such as this would either be successful, with an expected gross return of Rs 1,00,000, or unsuccessful, with an expected gross return of Rs 20,000. Similar products manufactured by the company have a record of being successful about 50 per cent of the time. The production and marketing costs of the new product are expected to be Rs 50,000.
   The company is considering whether to market this new product or to drop it. Before making its decision, a test marketing effort can be conducted at a cost of Rs 10,000. Based on past experience, test marketing results have been favourable about 70 per cent of the time. Furthermore, products favourably tested have been successful 80 per cent of the time. However, when the test marketing result has been unfavourable, the product has only been successful 30 per cent of the time. What course of action should the company pursue?
3. XYZ Company manufactures guaranteed tennis balls. At present, approximately 10 per cent of the tennis balls are defective. A defective ball leaving the factory costs the company Re 0.50 to honour its guarantee. Assume that all defective balls are returned. At a cost of Re 0.10 per ball, the company can conduct a test that always correctly identifies both good and bad tennis balls.
   (a) Draw a decision tree and determine the optimal course of action and its expected cost.
   (b) At what test cost should the company be indifferent to testing?
4. The Ore Mining Company is attempting to decide whether or not a certain piece of land should be purchased. The land costs Rs 3,00,000. If there are commercial ore deposits on the land, the estimated value of property is Rs 5,00,000. If no ore deposits exist, however, the property value is estimated at Rs 2,00,000. Before purchasing the land, the property can be cored at a cost of Rs 20,000. The coring will indicate if conditions are favourable or unfavourable for ore mining. If the coring report is favourable the probability of recoverable ore deposits on the land is 0.8. However, if the coring report is unfavourable then probability is only 0.2. Prior to obtaining any coring information, the management estimates that the odds of ore being present on the land is 50–

50. Management has also received coring reports on pieces of land similar to the ore in question and found that 60 per cent of the coring reports were favourable.

Construct a decision tree and determine whether the company should purchase the land, decline to purchase it, or take a coring test before making its decision. Specify the optimal course of action and EMV.

5. XYZ company, dealing with a newly invented telephone device, is faced with the problem of selecting out of the following available courses of action:
   (i) manufacture the device itself; or
   (ii) be paid on a royalty basis by another manufacturer; or
   (iii) sell the rights for its invention for a lump sum.

   The profit (in '000s Rs) that can be expected in each case and the probabilities associated with the level of sales are shown in the following table:

| *Outcome* | *Probability* | *Manufacture Itself* | *Royalties* | *Sell All Rights* |
|---|---|---|---|---|
| High sales | 0.1 | 75 | 35 | 15 |
| Medium sales | 0.3 | 25 | 20 | 15 |
| Low sales | 0.6 | – 10 | 10 | 15 |

   Represent the company's problem in the form of a decision tree. Further redraw the decision tree by introducing the following additional information:
   (a) If it manufactures itself, and the sales are medium or high, then the company has the opportunity of developing a new version of its telephone;
   (b) From past experience the company estimates that there is a 50 per cent chance of successful development,
   (c) The cost of development is Rs 15 and returns after deduction of development cost are Rs 30 and Rs 10 for high and medium sales, respectively.

6. An automobile owner faces the decision as to which deductible amount of comprehensive insurance coverage to select. Comprehensive coverage includes losses due to fire, vehicle theft, vandalism and natural calamites. The possible choices are zero deductible coverage for Rs 60 per year or Rs 50 deductible coverage for Rs 45 per year. (The owner pays the first Rs 50 of any loss of at least Rs 500.) Considering accidents covered by the comprehensive portion of the policy some of the owner's utility values are given below:

| | | | | | | |
|---|---|---|---|---|---|---|
| Amount (Rs) | : | – 95 | – 60 | – 50 | – 45 | 0 |
| Utility | : | 0.2 | 0.4 | 0.45 | 0.47 | 0.5 |

   (a) Sketch the owner's utility curve. Is the owner a risk avoider, an EMV'er or a risk taker?
   (b) On the basis of the utility curve drawn in part (a), should the owner take the zero deductible or the Rs 50 deductible comprehensive coverage?

7. Suppose a company has several independent investment opportunities each of which has an equal chance of gaining Rs 1,00,000 or losing Rs 60,000. What is the probability that the company will lose money on two such investments? On three such investments? On four such investments?

   If a company has a number of independent investment opportunities, in each of which the financial risk is relatively small compared to its overall asset position, why should the company try to maximize EMV, rather than expected utility?

8. An oil drilling company is considering the purchase of mineral rights on a property of Rs 100 lakh. The price includes tests to indicate whether the property has type A geological formation or type B geological formation. The company will be unable to tell the type of geological formation until the purchase is made. It is known, however, that 40 per cent of the land in this area has type A formation and 60 per cent type B formation. If the company decides to drill on the land it will cost Rs 200 lakh. If the company does drill it may hit an oil well, gas well or a dry hole. Drilling experience indicates that the probability of striking an oil well is 0.4 on type A and 0.1 on type B formation. Probability of hitting gas is 0.2 on type A and 0.3 on type B formation. The estimated discounted cash value from an oil well is Rs 1,000 lakh and from a gas well is Rs 500 lakh. This includes everything except cost of mineral rights and cost of drilling. Use the decision-tree approach and recommend whether the company should purchase the mineral rights?

9. The Sensual Cosmetic Co. has developed a new perfume which management feels has a tremendous potential. It not only interacts with the wearer's body chemistry to create a unique fragrance but is also especially long lasting. A total of Rs 10 lakh has already been spent on its development. Two marketing plans have been devised:
   (a) The first plan follows the company's usual policy of giving small samples of the new products when other items in the company's product lines are purchased and placing advertisements in women's magazines. The plan would cost Rs 5 lakh and it is believed that it might result in a high, moderate or low market response with probability of 0.2, 0.5 and 0.3, respectively. The net profit excluding development and promotion costs in these cases would be Rs 20 lakh, Rs 10 lakh and Rs 1 lakh, respectively. If it later appears that the market response is going to be low it would be possible to launch a TV advt. campaign. This would cost another Rs 7.5 lakh. It would change the market response to high or moderate as previously described but with probability of 0.5 each.
   (b) The second marketing plan is much more aggressive than the first. The emphasis would be heavily upon TV advertising. The total cost of this plan would be Rs 15 lakh, but the market response would be either excellent or good, with probabilities of 0.4 and 0.6, respectively. The profit excluding development and promotion costs would be Rs 30 lakh and Rs 25 lakh for the two outcomes.

   Advise on the sequence of strategy to be followed by the company.

10. The investment staff of TNC Bank is considering four investment proposals for a client: shares, bonds, real estate and saving certificates. These investments will be held for one year. The past data regarding the four proposals are given below:

    *Shares:* There is a 25 per cent chance that shares will decline by 10 per cent, a 30 per cent chance that they will remain stable and a 45 per cent chance that they will increase in value by 15 per cent. Also the shares under consideration do not pay any dividends.

    *Bonds:* These bonds stand a 40 per cent chance of increase in value by 5 per cent and 60 per cent chance of remaining stable and yield 12 per cent.

    *Real Estate:* This proposal has a 20 per cent chance of increasing 30 per cent in value, a 25 per cent chance of increasing 20 per cent in value, a 40 per cent chance of increasing 10 per cent in value, a 10 per cent chance of remaining stable and a 5 per cent chance of losing 5 per cent of its value.

    *Savings Certificates:* These certificates yield 8.5 per cent with certainty.

    Use a decision tree to structure the alternatives to the investment staff, and using the expected value criterion, choose the alternative with the highest expected value.

11. A company has developed a new product in its R&D laboratory. The company has the option of setting up production facility to market this product straightaway. If the product is successful, then over the three years expected product life, the returns will be Rs 120 lakh with a probability of 0.70. If the market does not respond favourably, then the returns will be only Rs 15 lakh, with a probability of 0.30.

The company is considering whether it should test-market this product by building a small pilot plant. The chance that the test market will yield favourable response is 0.80. If the test market gives a favourable response, then the chance of successful total market improves to 0.85.

If the test market give a poor response, then the chance of success in the total market is only 0.30.

As before, the returns from a successful market would be Rs 120 lakh and from an unsuccessful market only Rs 15 lakh. The installation cost to production is Rs 40 lakh and the cost of test-marketing the pilot plant is Rs 5 lakh. Using decision-tree analysis, draw a decision tree diagram, carry out necessary analysis to determine the optimal decisions.

[*Delhi Univ., MBA, 2002*]

## HINTS AND ANSWERS

1. The company should test market rather than drop the product. If test market result is positive, the company should market the product otherwise drop the product.
2. Test market should be followed; EMV = Rs 13,800
3. (a) Do not test, Re 0.05; (b) Re 0.05
4. Test: If favourable – purchase; If not – do not purchase; Rs 64,000
5. Royalties; IMV = Rs 15.5

   If $p$ is the probability of high sale, then

   (i) manufacture itself when $p > 0.24$

   (ii) paid on royalty basis when $0.07 < p < 0.24$

   (iii) sell all rights for lump sum when $p < 0.071$
6. (b) Rs 50 deductible
7. 0.25; 0.50; 0.3125
8. 

| | *Decision Point* | *Outcome* | *Probability* | *Conditional Value (Rs)* | *Expected Value* |
|---|---|---|---|---|---|
| $D_3$ | (i) Drill | Oil well | 0.1 | 1,000 | 100 |
| | | Dry hole | 0.6 | 0 | 0 |
| | | Gas | 0.3 | 500 | 150 |
| | | | | | 250 |
| | | | | Less: Cost | 200 |
| | | | | | 50 |
| | (ii) Do not drill | | | | 0 |
| | | | | | Total 50 |
| $D_2$ | (i) Drill | Oil well | 0.4 | 1,000 | 400 |
| | | Dry hole | 0.4 | 0 | 0 |
| | | Gas | 0.2 | 500 | 100 |
| | | | | | 500 |
| | | | | Less: Cost | 200 |
| | | | | | 300 |
| | (ii) Do not drill | | | | 0 |
| | | | | Total | 300 |
| $D_1$ | (i) Type A formation | | 0.4 | 300 | 120 |
| | (ii) Type B formation | | 0.6 | 50 | 30 |
| | | | | | 150 |
| | | | | Less: Cost | 100 |
| | | | | Total | 50 |

The company should purchase the mineral rights.

9. Aggressive Plan.

## CHAPTER SUMMARY

Decision theory provides a frame work for rational decision making in an environment of uncertainty and risk. In certain situations, a decision-maker needs to make either a single decision or a sequence of decisions (with additional information that may be available between decisions). A number of courses of action (strategies or decision alternatives) are available for each decision. The uncontrollable factors (also called states of nature) affect the payoff likely to be obtained from a course of action.